

Lab 3 Report

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Introduction:

The task for this lab is to design a circuit such that it takes a signal from a temperature sensor, and illuminates LEDs based on the signal. The requirements are that a green LED is lit for temperatures below 40 degrees and will turn off when above 40 degrees. A yellow LED is lit for temperatures above 40 degrees, and a red LED will be lit when the temperature measured rises above 100 degrees. It is also required that the red LED turns off when the temperature falls below 80 degrees, and the yellow LED turns off when the temperature falls below 30 degrees.

Design: Schmidt Triggers to control three LEDs

The first control circuit in this design is a Schmidt trigger. This circuit uses a reference voltage and feedback to switch between a high and low output state. The output of the trigger then controls the green and yellow LEDs. In this circuit, $V(TL)$ is 0.3V and $V(TH)$ is 0.4V. This is to control when the output is low or high based on $V(IN)$. The resistor ratio is set to 0.01, which is based on $V(CC)$, $V(TH)$, and $V(TL)$. $V(REF)$ is set to 0.347V. Essentially, when the input voltage is between 0V and 3V, the output is low, -5V. When the input voltage is greater than 4V, the output is high, +5V. These high and low voltages then are output to the yellow and green LEDs placed in parallel, with the green LED being oriented negatively towards $V(OUT)$ and the yellow LED being oriented positively towards $V(OUT)$.

The second control circuit takes the same $V(IN)$ as the first circuit, but has different values beyond that. The second circuit controls only the red LED. This control circuit has a $V(TL)$ of 0.8V and a $V(TH)$ of 1V, as well as a resistor ratio of 0.02. $V(REF)$ is set to 0.88V. This ensures that when $V(IN)$ is below 0.8V, the output is low, -5V. When the V_{in} is above 0.8V, the output is high, 5V. In reality, the red LED switches at 0.9V and not 0.8V. The reasoning for this is due to experimental error, however the error is unknown. Calculations show that the switching point should be 0.8V. The error could be due to using a physically incorrect resistor which alters the ratio.

Since the voltages are relatively low in this application there are no current limiting resistors for the LEDs in this circuit. The circuit performs consistently, some kind of small tweak must be made to the red control circuit for perfect performance.

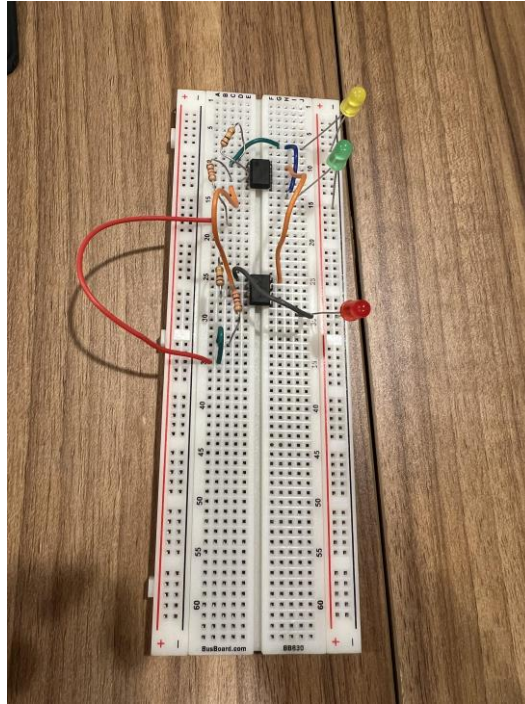


FIGURE 1: Temperature Indicator Circuit

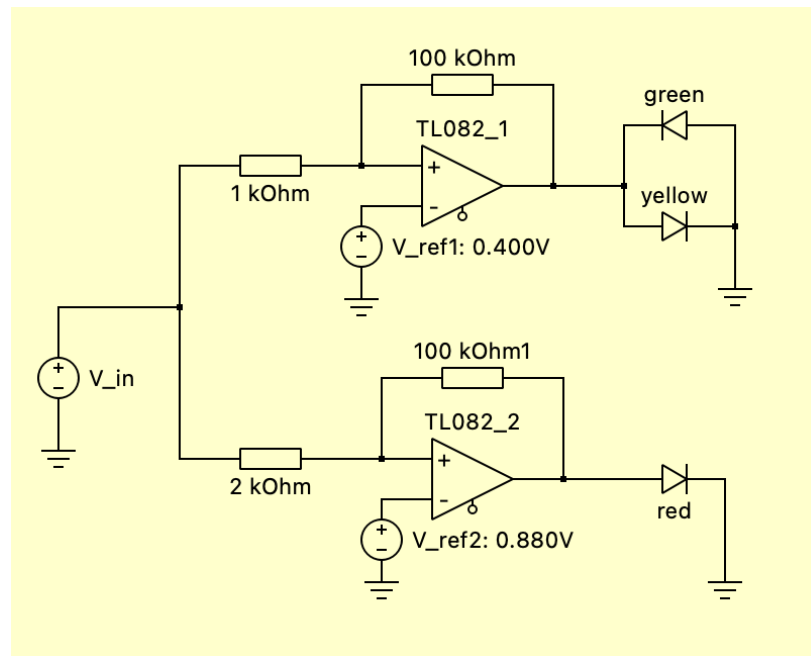


FIGURE 2: Schematic Design of Circuit

Conclusion:

In conclusion, this lab proved to be difficult. Many different configurations were tested, from 2 to 4 Op-Amps. After diving more into the theory, Schmidt Triggers were chosen as the

most straight forward option for this design. After figuring out how to calculate V_+ , V_- , $V_{(TH)}$, $V_{(TL)}$, (R_a/R_b) , and $V_{(REF)}$, these triggers were not complicated to implement.